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1) $M = \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -2 & 1 & 4 \end{bmatrix}$

Langkah 1 (Pengurangan) :

R_4 (baris 4) $= R_4 - R_3$ (dipilih karena mirip)

$= [(3-3) \ (-2-(-2)) \ (4-1) \ (4-0)]$

hasil matriks $= \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

Langkah 2 (Penjumlahan)

R_3 (baris 3) $= R_3 + R_2$

$= [(3+(-3)) \ (-2+2) \ (1+0) \ (0+3)]$

hasil matriks : $\begin{bmatrix} 2 & 3 & -1 & 1 \\ 0 & 0 & 1 & 3 \\ 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

Langkah 3 (Pertukaran)

R_2 ($R_2 \leftrightarrow R_3$)

hasil matriks : $\begin{bmatrix} 2 & 3 & -1 & 1 \\ 3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

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Langkah 4 (nol kan angka)

$(R_3 \leftarrow R_2 - \frac{3}{2} R_2)$

kolom 2 : $-2 - 3/2\,(3) = -2 - 9/2 = -13/2$

kolom 3 : $1 - 3/2\,(-1)) = 1 + 3/2 = 5/2$

kolom 4 : $0 - 3/2\,(1) = -3/2$

hasil matriks : $\begin{bmatrix} 2 & 3 & -1 & 1 \\ 0 & -13/2 & 5/2 & -3/2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ (matriks segitiga atas)

Determinan matriks : $|M| = -1 \times (2 \times -13/2 \times 1 \times 4)$

$|M| = -1 \times (-52)$

$= 52$

2) $M = \begin{bmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$

faktorkan angka 2 pada baris ke 2

$R_2 = \ldots \times 2 = 2 \quad = 1 \quad \ldots \times -2 = -2 \quad = (-1)$

hasil matriks : $\begin{bmatrix} 3 & 3 & 0 & 5 \\ 1 & 1 & 0 & -1 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$

kali baris 2 dan 3

$R_2 : [\,(1 \times 3)\ (1 \times 3)\ (0 \times 3)\ (-1 \times 3)\,]$

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hasil matriks : $\begin{bmatrix} 3 & 3 & 0 & 5 \\ 3 & 3 & 0 & -3 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$ 2

kurangi baris ke -1 dengan baris ke-2

$(R_1 - R_2)$: $[\ (3-3)\ (3-3)\ (0-0)\ (5-(-3))$

hasil matriks : $\begin{bmatrix} 0 & 0 & 0 & 8 \\ 1 & 1 & 0 & -1 \\ 4 & 1 & -3 & 0 \end{bmatrix}$ 2

baris 1 di pindah ke bawah

hasil matriks : $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \\ 0 & 0 & 0 & 8 \end{bmatrix}$ 2

$(R_2 \leftarrow R_2 - 4R_1)$: $(4-1(4))\ (1-1(1))\ (-3-1(4))\ (0-1(4))$

$(R_3 \leftarrow R_3 - 2R_1)$: $(2-1(2))\ (10-1(2))\ (3-1(2))\ (2-1(2))$

hasil matriks : $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & -3 & -3 & 4 \\ 0 & 8 & 3 & 4 \\ 0 & 0 & 0 & 8 \end{bmatrix}$ -2

kolom 3 : $3 + 8/3\ (-3)$: $3 - 8 = -5$

kolom 4 : $4 + 8/2\ (4)$: $4 + 32/3 = 44/3$

hasil matriks : $\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & -3 & -3 & 4 \\ 0 & 0 & -5 & 44/3 \\ 0 & 0 & 0 & 8 \end{bmatrix}$ -2

Determinan

$= |M| = -2 \times (1 \times -3 \times -5 \times 8)$

$= |M| = -2 \times (120)$

$= -240$

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